<u>DELIVER DCT</u>

1. <u>OVERVIEW:</u>

What is this use case about?

Answer:

Deliver DCT provides real-time and MTD turnover visibility across regions and CBFs by tracking sales performance against targets. It enables monitoring of sales pace, identification of potential turnover gaps, and focused interventions to optimize sales execution and month-end performance.

When is the report updated?

Answer:

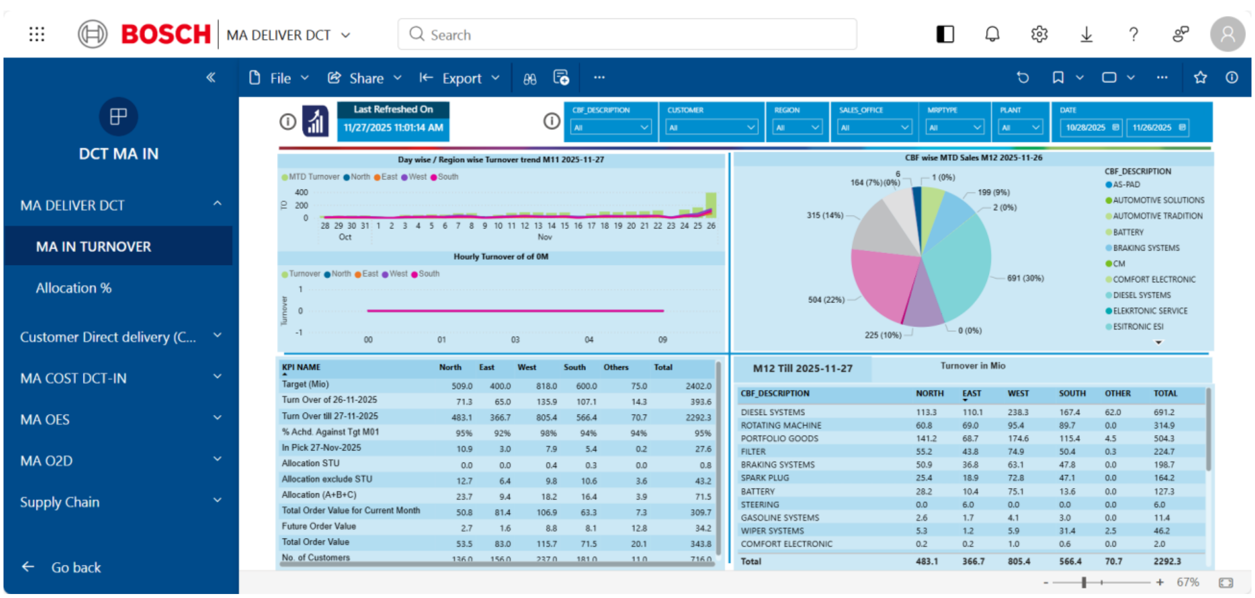
Refresh - Hourly, 0:00

KPI?

Answer: MTD Turnover

2. <u>USER DOCUMENTATION:</u>

This section details the key visual components and their interpretation within the Deliver DCT dashboard.



2.1 Dashboard Overview & Filters

The Deliver DCT dashboard offers a comprehensive view of turnover. Top filters (Customer, Region, Sales Office, Material Type, Plant, Date Range) allow granular data analysis.

2.1.1. Turnover Period Logic:

IAM Channel: Turnover month closes 4 days before the calendar month end (e.g., 27th for a 31st month).

2.1.2. Day-wise MTD Turnover Chart (Top Left):

Shows: Daily turnover progression for the current turnover month.

Purpose: Tracks sales pace against targets and identifies daily contributions, especially vital for month-end performance monitoring.

2.1.3. Hourly Turnover Trend (Bottom Left):

Shows: Real-time (hour-by-hour) daily turnover.

Purpose: Detects sudden changes, spikes, or invoicing delays, critically monitored between the 20th-27th for data refresh and dynamic sales tracking.

2.1.4. CBF-wise MTD Sales – Pie Chart (Top Right):

Shows: Total MTD turnover and percentage contribution of each product family.

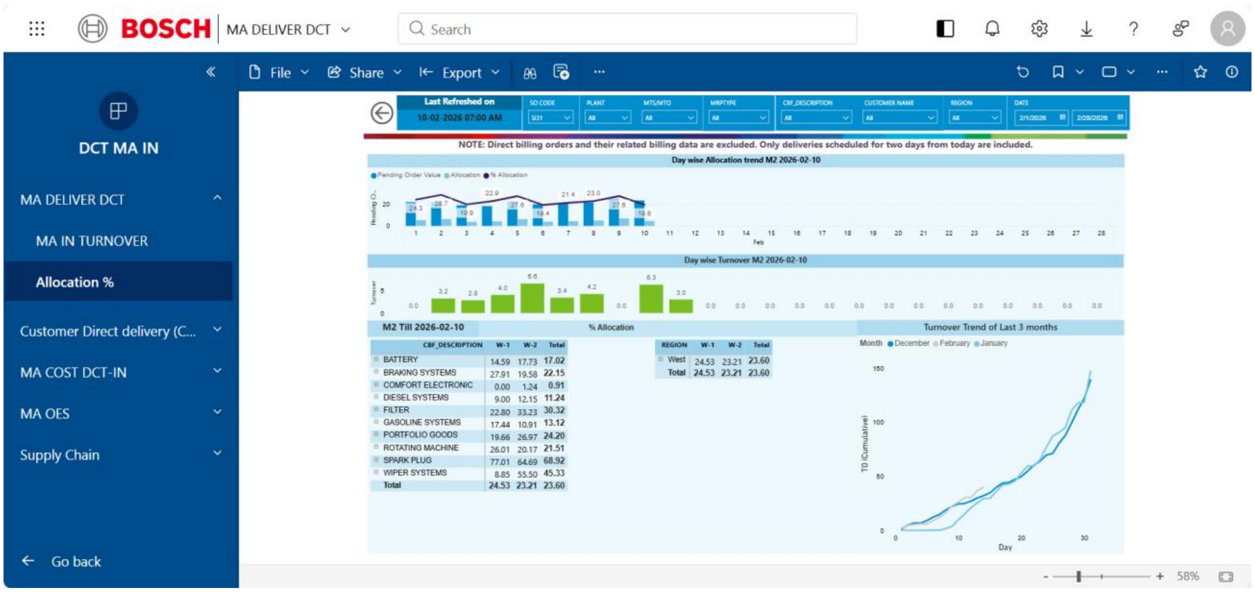
Purpose: Identifies top-performing product lines and areas needing sales focus.

2.1.5. Region-Wise Turnover Table (Bottom Right):

Shows: Turnover for each CBF across specific regions (North, East, West, South, Other), with grand totals.

Purpose: Allows sales leads to identify regional strengths and weaknesses for specific product lines, guiding targeted interventions.

2.2. This section provides critical visibility into the allocation process, showing pending order value and allocated quantities for upcoming deliveries.



2.2.1 Allocation Overview & Filters

The Allocation section of Deliver DCT shows pending order value and allocated quantity for near-term deliveries. Filters (TO ID-CODE, Plant, MTS/MTO, MRPTYPE, CBF_DESCRIPTION, CUSTOMER NAME, REGION, DATE) allow focused data filtering.

Note: Direct billing orders and related billing data are excluded. Only deliveries scheduled for the next two days are included.

2.2.2 Day-wise Allocation Trend (Top Section)

Shows: Day-wise Pending Order Value (bars), Day-wise Allocated Value (bars), % Allocation trend (line) for the selected month (e.g., Feb 2026)

Purpose: Provides day-wise visibility of pending order value, allocated value, and allocation percentage.

2.2.3 Day-wise Turnover Trend (Middle Section)

Shows: Actual day-wise turnover (bars) for the selected month (e.g., Feb 2026).

Purpose: Provides day-wise visibility of turnover values.

2.2.4 CBF-wise Allocation Table (Bottom Left)

Shows: Allocation percentage by CBF (Product Family), with week-wise values (W-1, W-2) and total allocation percentage.

Purpose: Provides CBF-wise visibility of allocation percentages.

2.2.5 Region-wise Allocation Table (Bottom Center)

Shows: Allocation percentage by region, with week-wise values (W-1, W-2) and total allocation percentage.

Purpose: Provides region-wise visibility of allocation percentages.

2.2.6 Turnover Trend – Last 3 Months (Bottom Right)

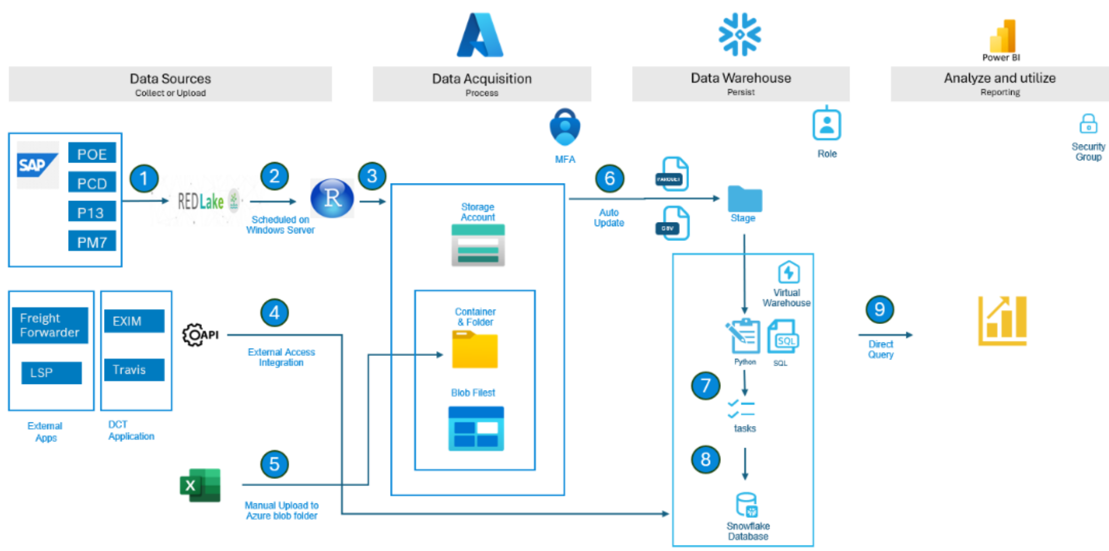
Shows: Cumulative turnover trend for the last three months.

Purpose: Provides visibility of turnover trends over the last three months.

3. <u>TECHNICAL DOCUMENTATION:</u>

This section briefly outlines the technical architecture underpinning the Bosch Deliver DCT system, highlighting its data flow and key components.

Overseas transportation architecture



3.1 Deliver DCT Technical Architecture Overview

The Deliver DCT system leverages a robust, cloud-enabled architecture to ensure reliable data acquisition, processing, storage, and reporting. It builds upon established Bosch data infrastructure to deliver actionable insights.

3.2 Key Architectural Phases

3.2.1 Data Sources:

SAP (POE, PCD, PL7): Core enterprise data from various SAP modules, covering procurement, commercial documents, and material management.

External Apps (Freight Forwarder, LSP, Travis, DCT App) & Excel: Integrates data from logistics partners, dedicated CDD applications, and allows for manual supplementary data uploads.

3.2.2 Data Acquisition:

RedLake & Windows Server: Extracts data from SAP via RedLake, with processes scheduled on Windows Servers for automation.

APIs & MFA: External application data is ingested via secure APIs. Multi-Factor Authentication (MFA) ensures secure access during data transfer.

Azure Storage Account: Data is stored in Azure Blob Storage (containers & folders) after initial acquisition, including manual uploads.

Auto Update: Automated processes ensure continuous data refresh within the storage account.

3.2.3 Data Warehouse:

Snowflake (Virtual Warehouse): Data is processed and persistently stored in Snowflake, a cloud data warehousing platform.

Staging: A temporary area for data validation and transformation.

Python & SQL: Used for complex data transformations, cleaning, and orchestration of data pipelines (tasks) within Snowflake.

3.2.4 Analyze & Utilize:

Power BI: The primary reporting tool, connecting directly to Snowflake via Direct Query. This enables near real-time dashboards and reports.

Security Groups: Access to Power BI reports is managed through security groups, ensuring data confidentiality and controlled access for authorized users.

This architecture ensures that the ABE system is scalable, secure, and provides timely, accurate data for proactive decision-making.

-> Power BI Dashboard Link: MA DELIVER DCT - Power BI

-> Snowflake Default Role Setup: [Snowflake Default Role Setup - 3S IT - End User Guide - Docupedia]

-> How to get Access: [DCT Access.docx]

-> Apply for the required role: [oneIDM]

QUESTIONS + ANSWERS

1. What is Deliver DCT?

ANSWER:

Deliver DCT is an operational dashboard used to monitor real-time and Month-To-Date (MTD) turnover performance across regions and CBFs.

Business Purpose:

- Track sales performance against targets
- Monitor sales pace
- Identify turnover gaps
- Support focused sales interventions
- Improve month-end execution performance

Primary KPI:

- MTD Turnover

2. What is the purpose of the Deliver DCT dashboard?

ANSWER:

The Deliver DCT dashboard provides operational visibility into turnover performance, allocation status, and regional sales execution.

Business Purpose:

- Monitor real-time turnover
- Track allocation performance
- Analyze regional and CBF-wise sales contribution
- Support sales execution monitoring
- Identify operational gaps affecting turnover achievement

The dashboard supports proactive operational and sales decision-making.

3. What is the main KPI in Deliver DCT?

ANSWER:

The primary KPI in Deliver DCT is:

- MTD Turnover (Month-To-Date Turnover)

Business Purpose:

- Measure sales execution performance
- Track turnover achievement against targets
- Monitor sales progression during the turnover month

4. How frequently is the Deliver DCT dashboard refreshed?

ANSWER:

The Deliver DCT dashboard is refreshed hourly starting from 0:00.

This enables near real-time visibility into:

- turnover progression
- allocation performance
- regional sales execution
- operational sales trends

5. What is IAM turnover period logic?

ANSWER:

In the IAM channel, the turnover month closes 4 days before the calendar month end.

Example:

- For a 31-day month, turnover closes on the 27th.

Business Purpose:

- Support operational billing closure
- Enable month-end sales consolidation
- Improve turnover reporting consistency

6. What is the purpose of the Day-wise MTD Turnover chart?

ANSWER:

The Day-wise MTD Turnover chart tracks daily turnover progression for the current turnover month.

Business Purpose:

- Monitor sales pace against targets
- Identify daily turnover contribution
- Support month-end execution tracking

This chart is especially important for monitoring turnover performance near month-end closure.

7. What is the purpose of the Hourly Turnover Trend?

ANSWER:

The Hourly Turnover Trend provides real-time hourly turnover visibility during the day.

Business Purpose:

- Detect sudden sales spikes
- Identify invoicing delays
- Monitor real-time turnover movement
- Support dynamic sales tracking

The trend is critically monitored between the 20th–27th of the turnover cycle.

8. What is CBF-wise MTD Sales analysis?

ANSWER:

CBF-wise MTD Sales analysis shows total Month-To-Date turnover and percentage contribution by product family.

Business Purpose:

- Identify top-performing product lines
- Detect low-performing CBFs
- Support focused sales intervention

The analysis helps prioritize operational sales focus areas.

9. What is the purpose of the Region-Wise Turnover table?

ANSWER:

The Region-Wise Turnover table provides turnover visibility across regions and CBFs.

Regions include:

- North
- East
- West
- South

- Other

Business Purpose:

- Identify regional strengths and weaknesses
- Compare product family performance by region
- Support targeted sales interventions

10. What is the purpose of the Allocation section in Deliver DCT?

ANSWER:

The Allocation section provides visibility into pending order value and allocated quantities for near-term deliveries.

Business Purpose:

- Monitor allocation performance
- Track pending order execution
- Improve delivery planning visibility
- Support short-term operational execution

Only deliveries scheduled for the next two days are included.

11. What is shown in the Day-wise Allocation Trend?

ANSWER:

The Day-wise Allocation Trend shows:

- Pending Order Value
- Allocated Value
- Allocation Percentage trend

Business Purpose:

- Monitor allocation efficiency
- Track pending execution value

- Analyze allocation progression during the month

12. What is the purpose of the Day-wise Turnover Trend?

ANSWER:

The Day-wise Turnover Trend provides visibility into actual daily turnover values for the selected month.

Business Purpose:

- Monitor daily sales execution
- Track turnover consistency
- Support operational sales monitoring

13. What is the purpose of the CBF-wise Allocation table?

ANSWER

The CBF-wise Allocation table provides allocation percentage visibility by product family (CBF).

The analysis includes:

- week-wise allocation values
- total allocation percentage

Business Purpose:

- Monitor allocation efficiency by product family
- Identify allocation gaps
- Support operational prioritization

14. What is the purpose of the Region-wise Allocation table?

ANSWER:

The Region-wise Allocation table shows allocation percentage by region with weekly allocation visibility.

Business Purpose:

- Monitor regional allocation performance
- Identify regional execution gaps
- Support targeted operational intervention

15. What is the purpose of the Turnover Trend – Last 3 Months analysis?

ANSWER:

The Turnover Trend – Last 3 Months analysis provides cumulative turnover visibility across the previous three months.

Business Purpose:

- Monitor turnover trends
- Analyze sales consistency
- Support operational performance comparison

16. Why is hourly turnover monitoring operationally important?

ANSWER:

Hourly turnover monitoring enables real-time visibility into invoicing and sales execution performance.

Operational Importance:

- Detect invoicing delays quickly
- Monitor sudden turnover changes
- Support faster corrective actions
- Improve month-end execution responsiveness

This is especially critical during the turnover closure period.

17. Why is allocation visibility important in delivery operations?

ANSWER:

Allocation visibility helps operational teams track whether pending customer orders are properly allocated for delivery execution.

Operational Importance:

- Improve delivery planning
- Reduce execution delays
- Monitor pending order risks
- Improve short-term fulfillment visibility

Poor allocation visibility can impact turnover achievement and delivery performance.

18. Why are turnover gaps operationally risky?

ANSWER:

Turnover gaps indicate a mismatch between sales targets and actual execution performance.

Operational Impact:

- Reduced sales achievement
- Lower month-end turnover performance
- Increased pressure on late-cycle invoicing
- Higher execution risk during turnover closure

Deliver DCT helps teams identify and respond to such gaps proactively.

19. What are the main data sources used in Deliver DCT?

ANSWER:

Deliver DCT integrates data from multiple enterprise and logistics systems.

Primary data sources:

- SAP systems (POE, PCD, PL7)
- Freight Forwarder applications
- LSP systems
- Travis
- DCT App
- Excel uploads

These systems support turnover tracking, allocation visibility, and operational reporting.

20. What is the role of Snowflake and Power BI in Deliver DCT?

ANSWER:

Snowflake acts as the cloud data warehouse platform, while Power BI serves as the reporting and visualization layer.

Snowflake Responsibilities:

- Data storage
- Data transformation
- Pipeline orchestration

Power BI Responsibilities:

- Near real-time dashboard reporting
- Operational turnover visibility
- Sales and allocation monitoring

Power BI connects directly to Snowflake using Direct Query.

21. Required Roles to Access DCT and Snowflake?

ANSWER:

i. IDM2BCD_SNOWFLAKE_PRXDMLB_BOSCH_DCTINDIA_DEV_GS_DEVELOPER

ii. IDM2BCD_SNOWFLAKE_PRXDMLB_BOSCH_DCTINDIA_DEV_MA_DEVELOPER